Victorine Younang, Ph.D.

Data Scientist / Machine Learning Practitioner — Ph.D., Computer Science

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SUMMARY

Data Scientist (Ph.D., Computer Science) with 5+ years of applied experience engineering end-to-end predictive models, scalable ETL pipelines, and NLP systems. Translates messy, unstructured data into measurable product and business impact — e.g., a recommender driving a +33% lift in session time, and clinical models targeting 30-day readmissions. Combines a Big-Data master's foundation (Hadoop, PySpark, Elasticsearch) with PhD-level rigor (Bayesian inference, time-series forecasting, experimentation).

TECHNICAL SKILLS

- **Statistics & Modeling:** predictive modeling, Bayesian inference, VAR/ARIMA time-series forecasting, causal inference, A/B testing, experimental design, XGBoost/LightGBM, scikit-learn
- **Data Engineering:** SQL, PySpark, Apache Spark, Hadoop/HDFS, Kudu, Elasticsearch, Impala, ETL pipelines; Snowflake, SAP HANA
- **ML / NLP:** PyTorch, Hugging Face, transformers (BERT/BioBERT/ClinicalBERT), NER, recommender systems (Node2Vec), GNNs; SHAP/LIME explainability
- **Deployment & BI:** Docker, MLflow, Kibana/BI dashboards; Python, R

EXPERIENCE

Oakland University | Data Scientist & ML Researcher

Michigan, USA · Sep 2020 – Apr 2026

- **Statistical Modeling & Forecasting:** Vector AutoRegression (VAR) for dynamic demand forecasting; resource-allocation policies that extended IoT network lifetime by 20% at 95% request fulfillment.
- **Imbalanced Data & Classification:** Graph Attention Network + MLP to detect structural anomalies in small-data, highly imbalanced regimes; variant-aware stratified sampling → F1 0.94 with zero-shot generalization.
- **Evaluation & Metric Design:** three custom evaluation metrics (syntactic validity, semantic match, value consistency) to rigorously quantify generative-model accuracy.

Health Alliance Plan (HAP) | Data Scientist (Predictive Analytics)

Detroit, MI · Summer 2025

- **Predictive Modeling & Business Impact:** end-to-end ML pipeline predicting 30-day ER readmissions, identifying high-risk members for preventative care and driving operational cost savings.
- **Advanced Data Engineering:** extracted unstructured clinical notes + structured claims from Snowflake via advanced SQL and Python text-preprocessing.
- **Supervised Learning & Optimization:** benchmarked XGBoost / LightGBM against deep models, tuning for Recall / Sensitivity on critical risk identification.
- **Stakeholder Analytics:** translated ROC-AUC and SHAP values into cost-saving estimates for senior analytics managers and clinical directors.

Rec Room | Data Scientist (Algorithms & Product Analytics)

Seattle, WA · Summer 2023

- **Product Recommender Systems:** structural content-recommendation algorithm (Node2Vec, random walk) over a bipartite graph of 1,000+ environments to personalize content.
- **A/B Testing & Business Impact:** led online evaluation of the recommender, driving a statistically significant +33% increase in average session time.
- **Pipeline Optimization:** refactored the Python inference pipeline, –10% time-to-recommendation latency, improving the end-user experience.

EDF (Électricité de France) | Data Scientist (Time-Series & Forecasting)

Paris, France · Summer 2021

- **High-Frequency Data Engineering:** end-to-end SQL / PySpark pipelines ingesting streaming telemetry from thousands of smart meters, SCADA systems, and satellite weather feeds.
- **Predictive Grid Analytics:** probabilistic deep-learning pipeline forecasting short-term load + renewable generation → optimized reserve margins and network stability.
- **Advanced Feature Engineering:** spatial/temporal embeddings from cyclical-time constraints and physical substation adjacency matrices.
- **End-to-End MLOps:** Dockerized preprocessing + models with MLflow tracking; presented Pinball Loss / CRPS validation to senior R&D stakeholders.

EDF (Électricité de France) | Data Scientist & Big Data Consultant

Paris, France · 2019–2020

- **Large-Scale Data Processing:** managed the big-data architecture for a consent platform scaling to millions of customer profiles on distributed Hadoop clusters.
- **Data Quality & Governance:** automated data-sanitization and validation pipelines enforcing GDPR compliance and reducing downstream reporting errors by 30%.
- **Cross-Functional Technical Leadership:** technical-functional lead translating regulatory requirements into deterministic data-engineering rules.

Orange Telecom | Data Scientist (Predictive Analytics & BI)
Paris, France & Brussels, Belgium · 2017–2019

- **Predictive Churn & Capacity Forecasting:** multivariate time-series and regression models in R forecasting network capacity and customer churn → proactive retention and scaling.
- **End-to-End ETL:** scalable Python / SQL pipelines loading massive XML customer/network data into Apache Kudu and Hadoop, cutting data-preparation time.
- **Business Intelligence:** interactive dashboards (Kibana, TIME Matrix) giving executives real-time visibility into network health and app-portfolio performance.
- **Process Optimization:** mapped and optimized the HUB DATA France application architecture, cutting cross-functional query-setup overhead by 15%.

SELECTED PROJECTS

- **RepoSage — Agentic, execution-verified RAG** (LangGraph, vLLM, Qdrant, DPO): dual-source code-&-docs RAG that generates, executes in a sandbox, and self-corrects against a library's own tests; DPO alignment from execution-feedback preferences.
- **NLP at Scale — BERT stance/sentiment classifier:** scraped 19,000+ posts; fine-tuned BERT to 96% accuracy with temporal topic modeling (Springer LNCS).
- **DALKIA Energy-Consumption Forecasting** (SAP HANA, SAP Predictive Analysis Library): predicted and visualized client energy consumption to drive optimization.

EDUCATION

- **Ph.D., Computer Science** — Oakland University, Michigan (Sep 2020 – Apr 2026; GPA 3.96). Dissertation: *From Perceived Trust to Provable Security: Design, Optimization & Verification of Resource-Constrained IoT Ecosystems.*
- **M.Sc. Computer Engineering & B.Sc. Electronic & Computer Engineering** — ESIGELEC, France (2016–2019; GPA 3.87). *Big Data for Digital Transformation.*

SELECTED PUBLICATIONS

- V. Younang et al. "Security Risk Assessment with Bayesian Attack Graphs and Complex Probabilities." *IEEE TDSC*, 2025.
- V. Younang et al. "Extending IoT Devices' Network Lifetime." *IEEE TSC*, 2025.
- V. Younang et al. "Comparative Analysis of Users' Concerns and Government Policies on Autonomous Vehicles." *Springer ICIOT / LNCS*, 2022.
- V. Younang et al. "CipherGuard: Reference-Free Detection of Backdoors in Cipher Implementations." *Under review, NDSS 2027.*
- V. Younang et al. "Tiered Isomorphic Alignment: Curriculum Fine-Tuning across Six LLMs." *Under review, AAAI 2027.*
- V. Younang et al. "CryptoGraph: Learning Security Properties of Block Ciphers with Graph ML." *In prep (resubmission), IEEE TIFS.*

CERTIFICATIONS

- **AWS Certified AI Practitioner (AIF-C01)** — expected Nov 2026; AWS AI/ML stack incl. Amazon Bedrock and SageMaker.
- **NVIDIA-Certified Associate: Generative AI & LLMs (NCA-GENL)** — expected Dec 2026; transformers, prompt engineering, fine-tuning, NeMo / Triton / TensorRT.
- **OWASP Top 10 for LLM Applications** — de-facto LLM-security framework (prompt injection, insecure output handling, model DoS, supply-chain), demonstrated in RepoSage.